

GIitHub Links

- Lab 1 (Data Collection API) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/jupyter-labs-spacex-data-collection-api.ipynb
- Lab 2 (Web Scraping) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/jupyter-labs-webscraping.ipynb
- Lab 3 (Data Wrangling) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb
- Lab 4 (EDA with SQL) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb
- Lab 5 (EDA with Visualization) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/edadataviz.ipynb
- Lab 6 (Folium Map) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/lab_jupyter_launch_site_location.ipynb
- Lab 7 (Plotly Dash App) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/spacex-dash-app.py
- Lab 8 (Machine Learning Prediction) https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/blob/main/SpaceX_Machine%20Learning%20Prediction.ipynb



Applied Data Science Capstone

SALAH KASMO

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
- Innovative Insights & Business Impact
- Conclusion
- Appendix



EXECUTIVE SUMMARY

- **Project Overview & Goal**

Predict Falcon 9 landing success to optimize launch pricing and market bidding.

- **Data Processing**

- Extracted records via SpaceX API and Wikipedia web scraping.
- Labeled landing outcomes (Success =1, Failure = 0) and handled missing data.
- Encoded categorical features (Site, Orbit, Payload) for model readiness.

- **Exploratory Analysis (EDA)**

Identified success patterns by site and payload using SQL, Pandas, and Plotly Dash.

- **Predictive Modeling**

Trained and evaluated Logistic Regression, SVM, Decision Tree, and KNN.

- **Key Findings**

Decision Tree achieved top test accuracy (~83.3%), enabling reliable recovery predictions.

INTRODUCTION

- **SpaceX & Commercial Spaceflight Era**

SpaceX has revolutionized commercial space travel by drastically reducing launch costs compared to traditional aerospace contractors.

- **The Key Innovation: First-Stage Reusability**

The first-stage booster represents the largest portion of rocket manufacturing costs; recovering and reusing it is the key reason SpaceX charges around \$62 million per launch instead of \$165+ million.

- **Business & Economic Motivation**

Accurately determining booster landing success allows us to estimate the actual mission cost and financial risk for any given launch.

- **Primary Project Objectives**

Predict Falcon 9 booster landings using machine learning to accurately estimate launch costs and improve bidding competitiveness.



METHODOLOGY



- Data Collection and Data Wrangling Methodology
- EDA and Interactive Visual Analytics Methodology
- Predictive Analysis Methodology



Data Collection and Data Wrangling Methodology

- Data Acquisition

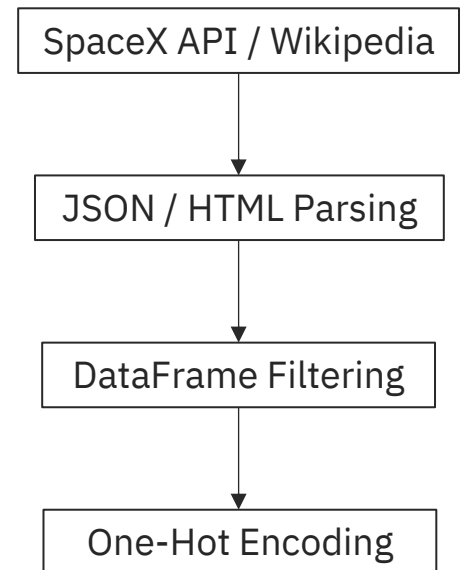
- Retrieved launch historical data using the SpaceX REST API (`/v4/launches`).
- Scraped supplementary Falcon 9 launch records from Wikipedia using *BeautifulSoup*.

- Data Cleansing & Filtering

- Filtered out Falcon 1 launches to focus exclusively on Falcon 9 missions (90 flights).
- Imputed missing values in payload mass using column mean values.

- Feature Engineering

- Created binary target labels (Class: \$1\$ for successful landing, \$0\$ for failure/uncontrolled).
- One-hot encoded categorical features (Launch Site, Orbit, and boosters) to prepare numeric matrices.



EDA and Interactive Visual Analytics Methodology

- Exploratory Data Analysis (EDA)
 - Queried operational datasets via SQL to determine success rates per site and payload distributions.
 - Computed flight patterns, trends over launch years, and success rate progression using Pandas and Seaborn.
- Interactive Mapping (Folium)
 - Marked launch sites, proximities to coastlines, railways, and highways using marker clusters.
 - Measured distances from launch pads to infrastructure to analyze geographical safety parameters.
- Dashboard Analytics (Plotly Dash)
 - Built an interactive dashboard with dropdowns to monitor site-specific success breakdowns via pie charts.
 - Implemented range sliders to examine payload mass correlations with booster recoveries on scatter plots.



Predictive Analysis Methodology

- Data Standardization & Splitting
 - Standardized input features using *StandardScaler* from Scikit-Learn.
 - Split data into training and test sets (80% train, 20% test) using *train_test_split*.
- Model Training & Hyperparameter Tuning
 - Evaluated four classification algorithms: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
 - Optimized model parameters across all classifiers using *GridSearchCV* with 10-fold cross-validation.
- Evaluation & Selection
 - Compared test accuracy scores and generated confusion matrices to select the best-performing model for deployment.

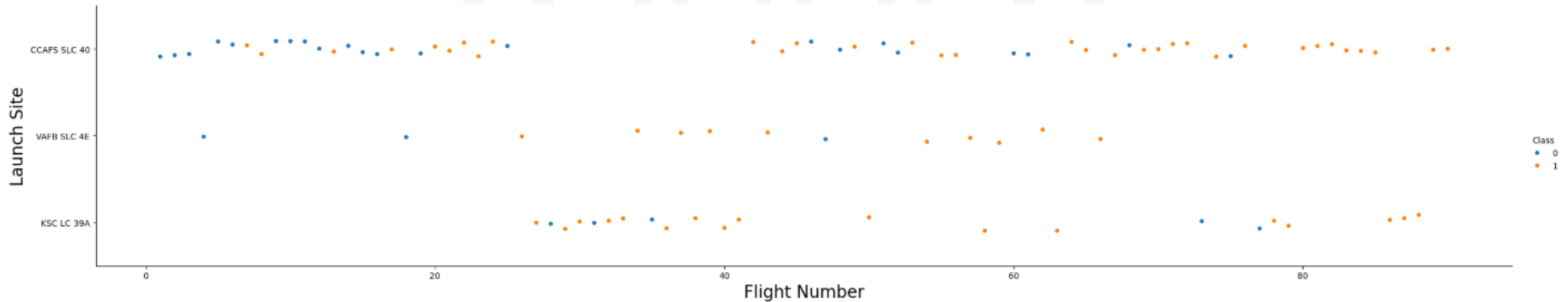


RESULTS

- EDA with Visualization Results
- EDA with SQL Results
- Interactive Map Results using Folium
- Plotly Dash Dashboard Results
- Predictive Analysis (Classification) Results

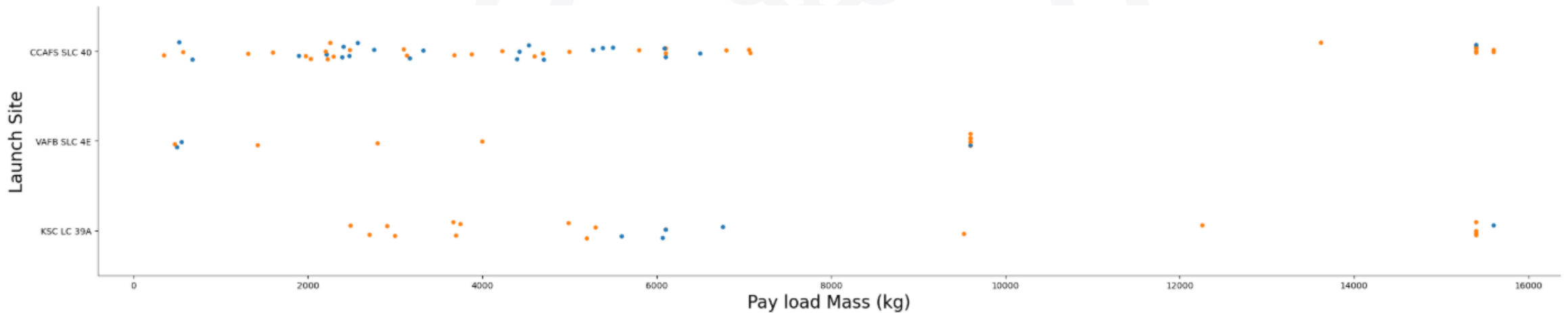
EDA with Visualization Results - 1

- Launch Site Success Rates:
 - KSC LC-39A achieved the highest overall success rate among all launch facilities.
 - CCAFS LC-40 showed a lower initial success rate due to hosting early experimental flights.



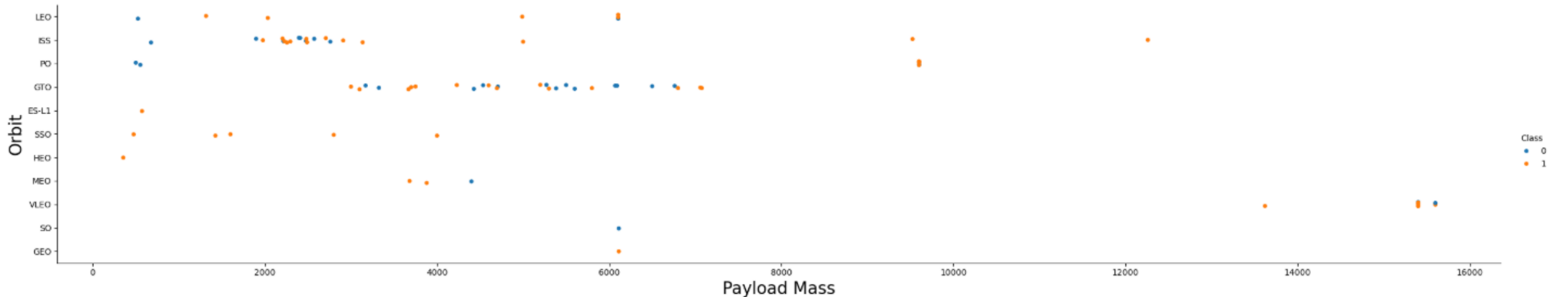
EDA with Visualization Results - 2

- Payload Mass & Success:
 - Light-to-medium payloads correlated strongly with higher booster recovery rates.



EDA with Visualization Results - 3

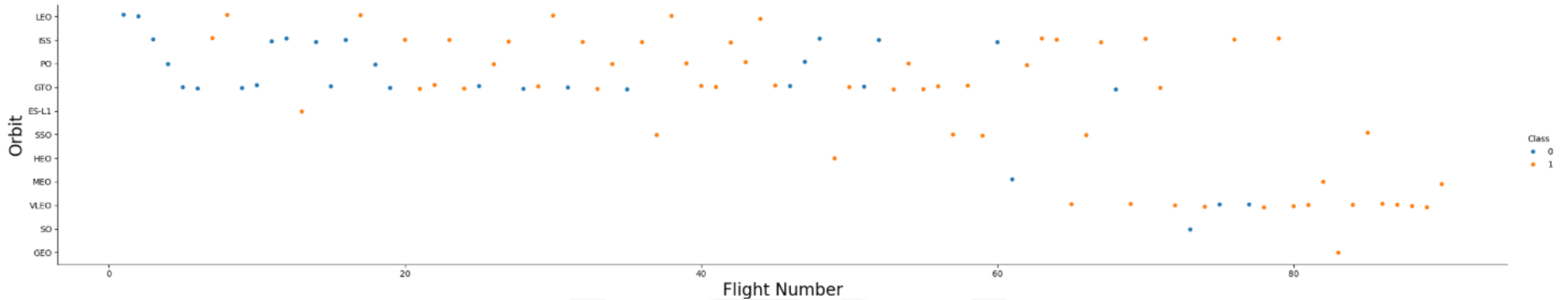
- Payload Mass & Success:
 - Heavy payloads (>8,000 kg, especially for Polar, LEO, and ISS orbits) showed higher landing success rates.



EDA with Visualization Results - 4

- Orbit Type & Success:

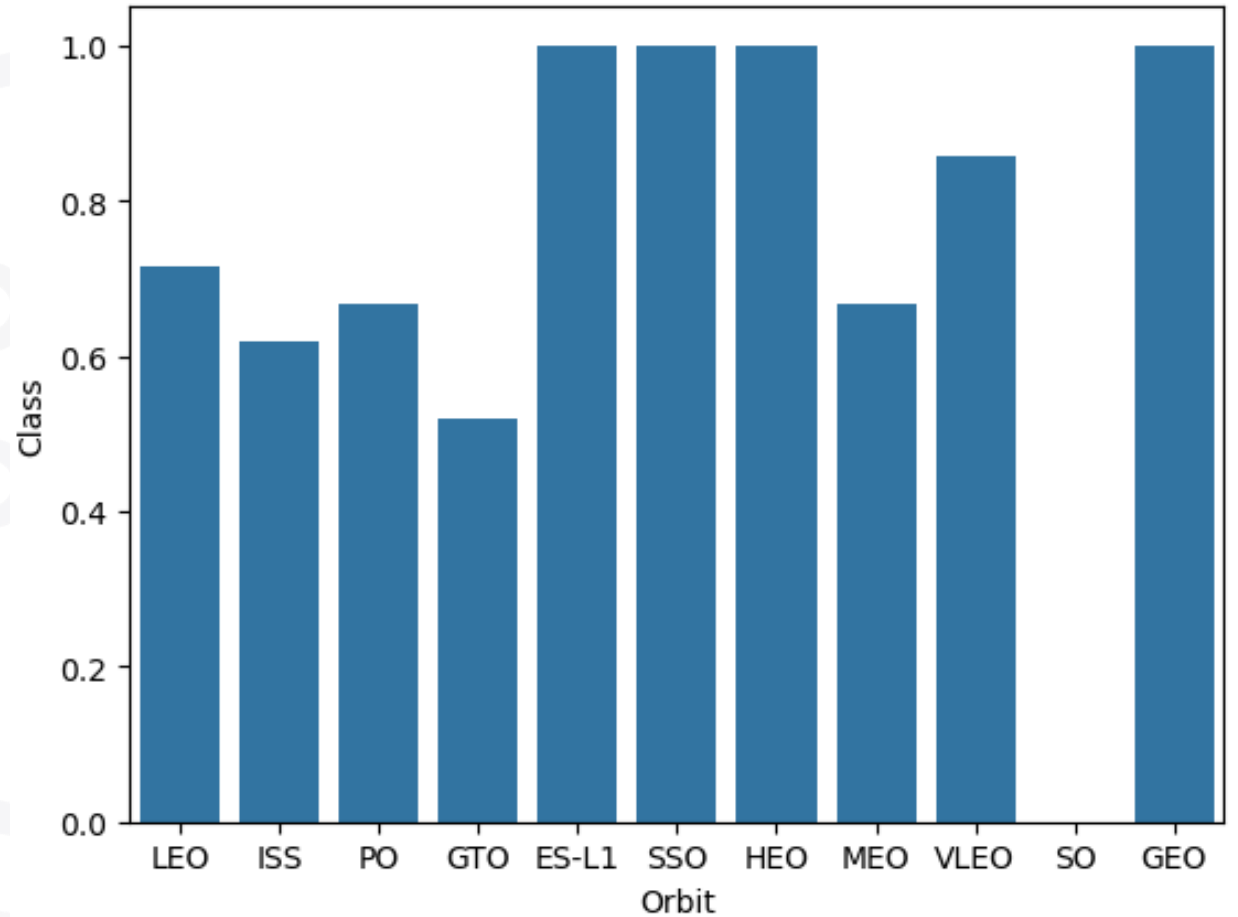
- The success landing of LEO orbit seems to have a relation with the flight number, while it has no relationship for GTO orbit .



EDA with Visualization Results - 5

- Orbit Type & Success:

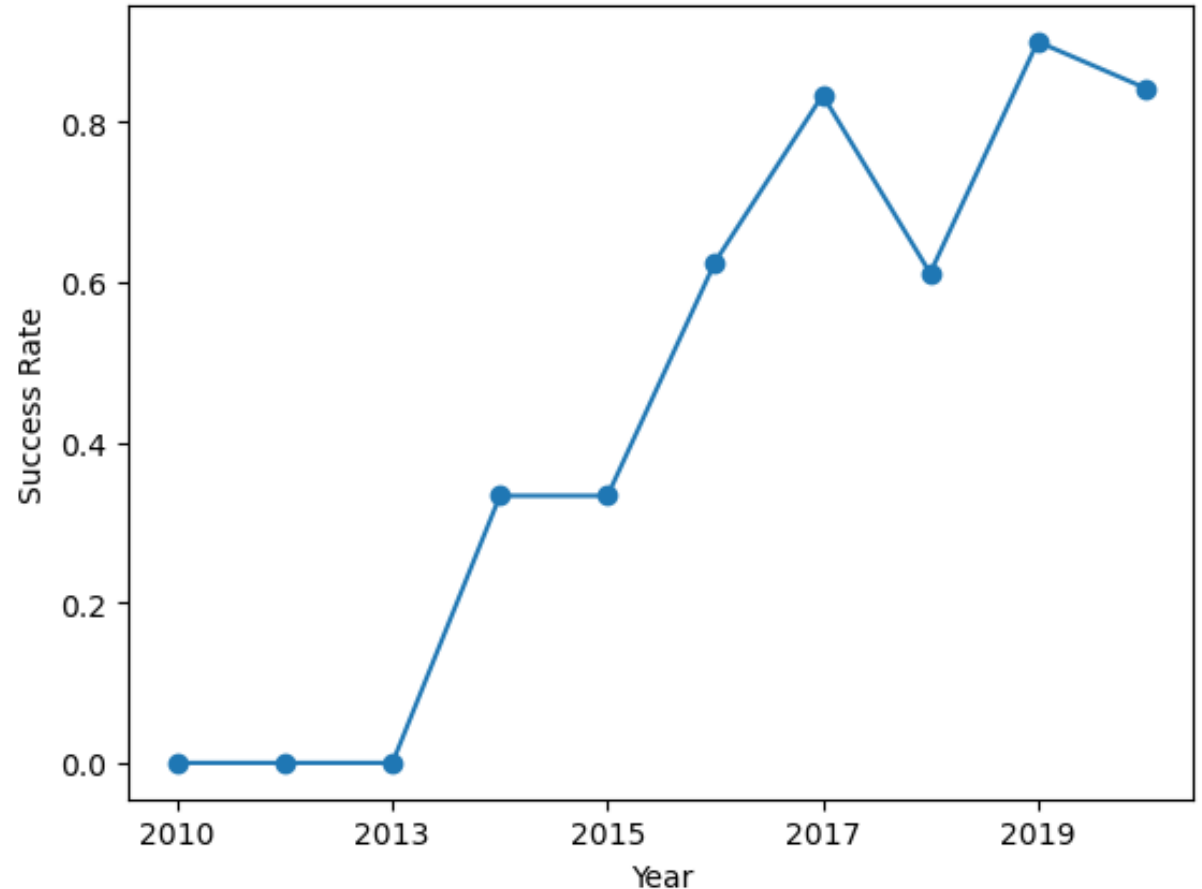
The orbits ES-L1, SSO, HEO and GEO have the highest success rates.



EDA with Visualization Results - 6

- Flight Progression:

Annual success rates improved consistently after 2015 with the deployment of advanced booster variants.



EDA with SQL Results -1

- Site Names & Missions:
- Identified all four launch sites and aggregated number of successful and failure mission outcomes (Success 100, Failure 1).

```
statement = 'SELECT DISTINCT Launch_Site FROM SPACEXTBL'  
cur.execute(statement)  
  
output=cur.fetchall()  
print (output)
```

```
[('CCAFS LC-40',), ('VAFB SLC-4E',), ('KSC LC-39A',), ('CCAFS SLC-40',)]
```

```
statement = '''SELECT Mission_Outcome, COUNT(*)  
                FROM SPACEXTBL  
                GROUP BY (Mission_Outcome)'''  
cur.execute(statement)  
  
output=cur.fetchall()  
print (output)
```

```
[('Failure (in flight)', 1), ('Success', 98), ('Success ', 1), ('Success (payload status unclear)', 1)]
```

EDA with SQL Results - 2

- Payload Aggregations:
 - The total payload mass carried by boosters launched by NASA (CRS) is 45596 kg.

```
statement = 'SELECT SUM (PAYLOAD_MASS_KG_) FROM SPACEXTBL WHERE Customer= "NASA (CRS)"'
cur.execute(statement)

output=cur.fetchall()
print (output)
```

```
[(45596,)]
```

- Average payload mass carried by booster version F9 v1.1 is 2928.4 kg.

```
statement = 'SELECT AVG (PAYLOAD_MASS_KG_) FROM SPACEXTBL WHERE Booster_Version= "F9 v1.1"'
cur.execute(statement)

output=cur.fetchall()
print (output)
```

```
[(2928.4,)]
```



EDA with SQL Results - 3

- Successful Landing Outcomes:
 - Documented the first successful landing outcome in ground pad was achieved in 2015.

```
statement = 'SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome= "Success (ground pad)"'
cur.execute(statement)

output=cur.fetchall()
print (output)
```

```
[('2015-12-22',)]
```

- The highest successful landing outcome is in drone ship.

```
statement = '''SELECT Landing_Outcome, count(*) as counts
FROM SPACEXTBL
where Date between '2010-06-04' AND '2017-03-20'
group by (Landing_Outcome)
order by (counts) desc
'''

cur.execute(statement)

output=cur.fetchall()
print (output)
```

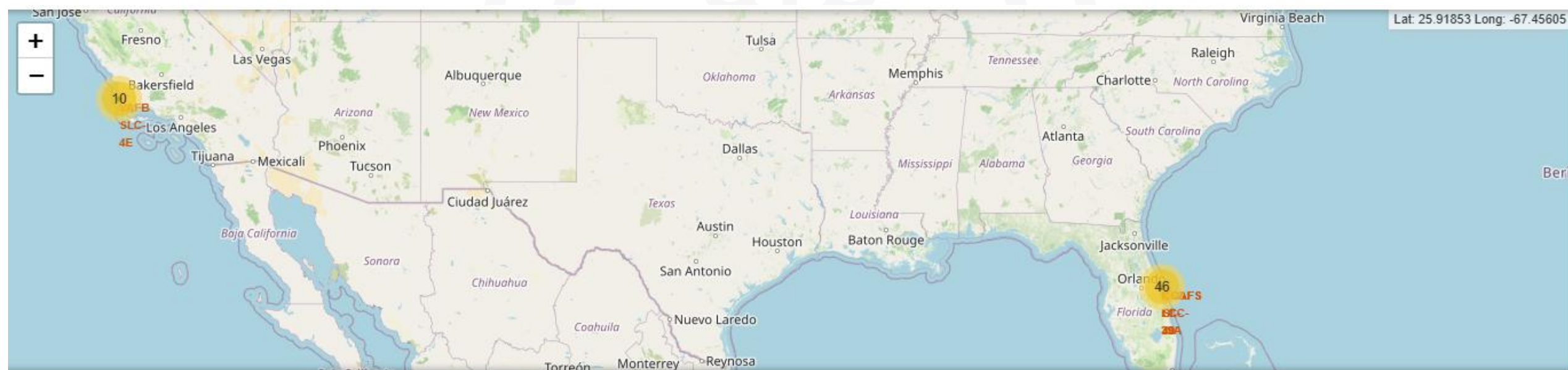
```
[('No attempt', 10), ('Success (drone ship)', 5), ('Failure (drone ship)', 5),
('und (drone ship)', 1)]
```



Interactive Map Results using Folium - 1

- Proximity to Water Bodies:

All launch complexes are situated along coastal waters to ensure safe oceanic flight paths away from populated zones.



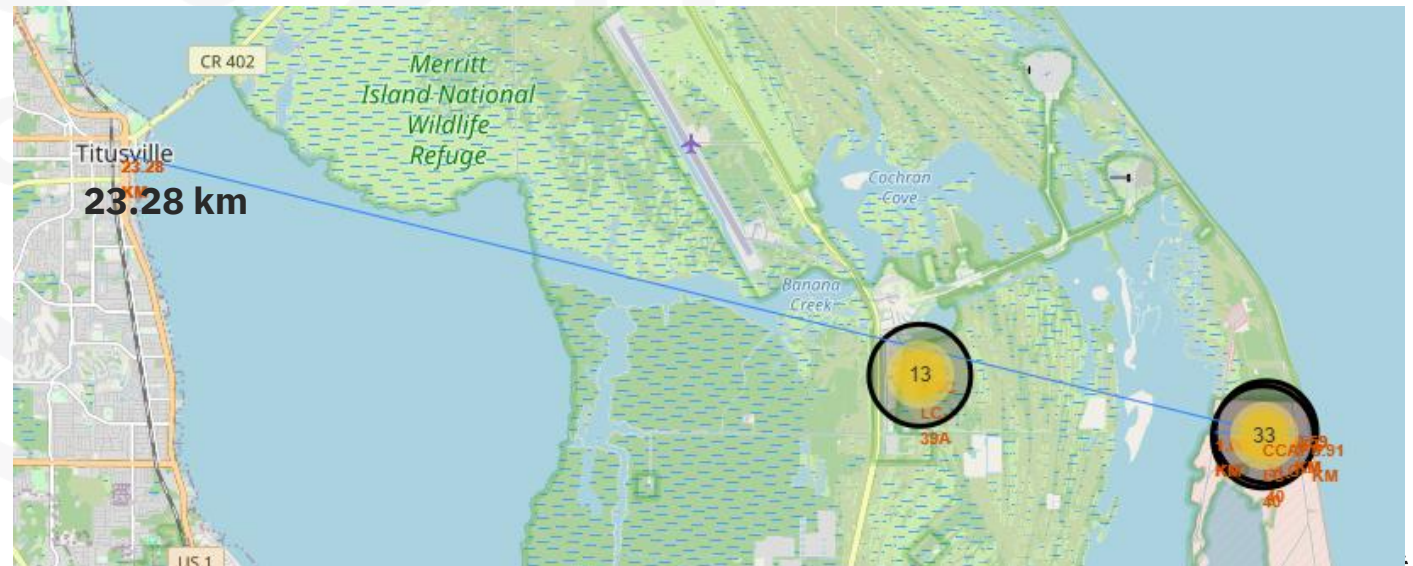
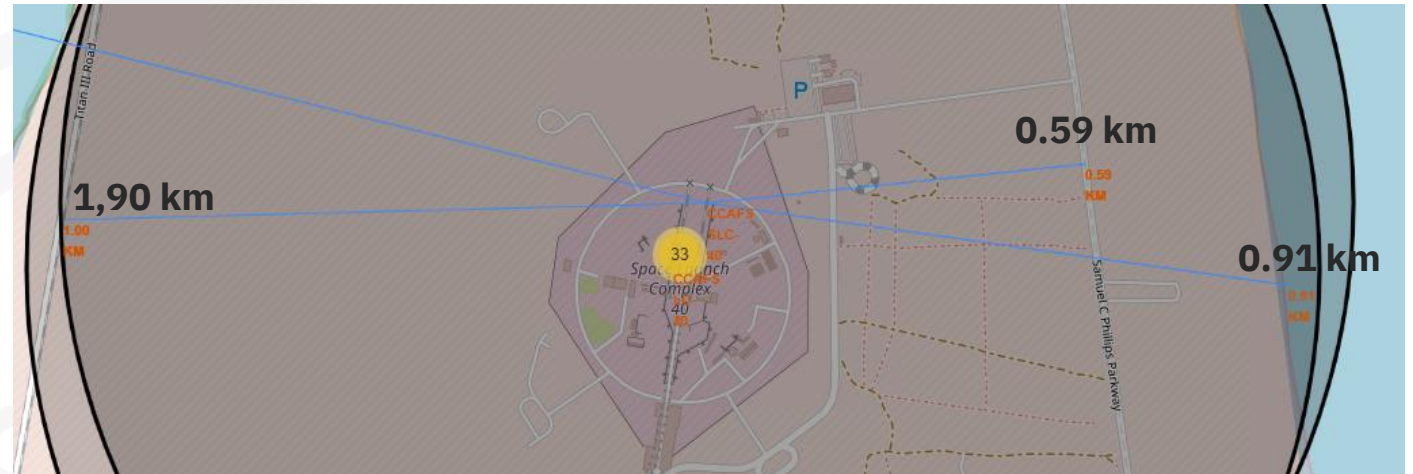
Interactive Map Results using Folium - 2

- Logistics & Infrastructure:

Launch pads are strategically located adjacent to major railways and highways to facilitate heavy rocket transport.

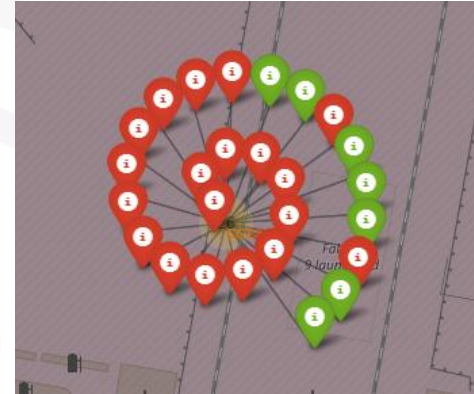
- Safety Distances:

Launch complexes maintain critical safety buffer distances from major residential areas and cities.

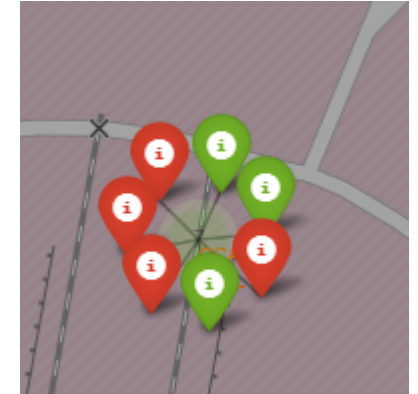


Interactive Map Results using Folium - 3

- KSC LC-39A: 13 launches (76.9% Success).
- CCAFS SLC-40: 33 launches (57.1% Success).
- VAFB SLC-4E: 10 launches (40.0% Success).
- CCAFS LC-40: 26 launches (73.1% Success).



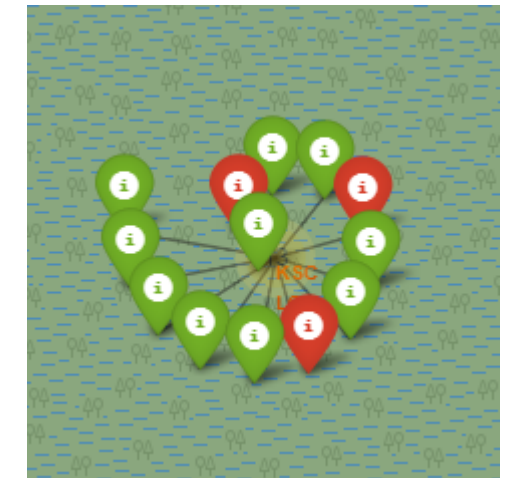
CCAFS LC-40



CCAFS SLC-40



VAFB SLC-4E



KSC LC-39A

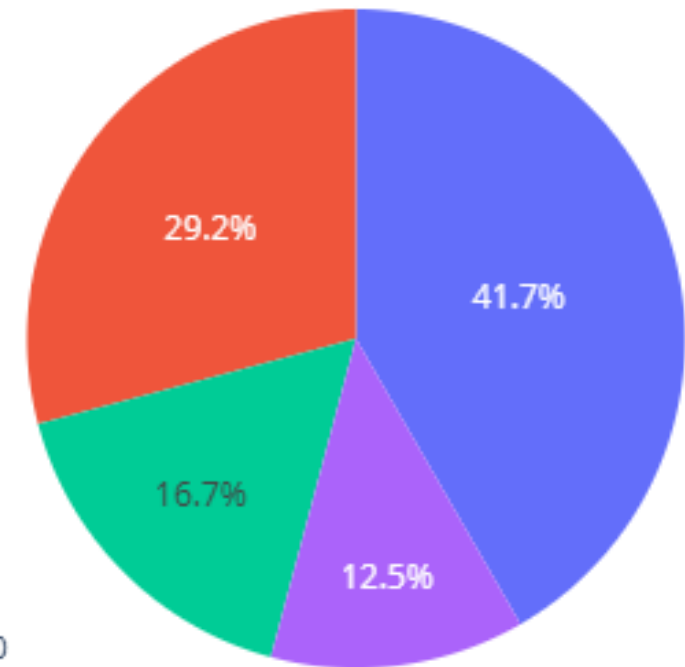
Plotly Dash Dashboard Results - 1

- The GitHub link of Dashboard's snap screens

https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/tree/main/Dash_snapscreen

- Interactive Filtering:

The interactive pie chart provided instant breakdowns of success vs. failure rates across individual launch sites.



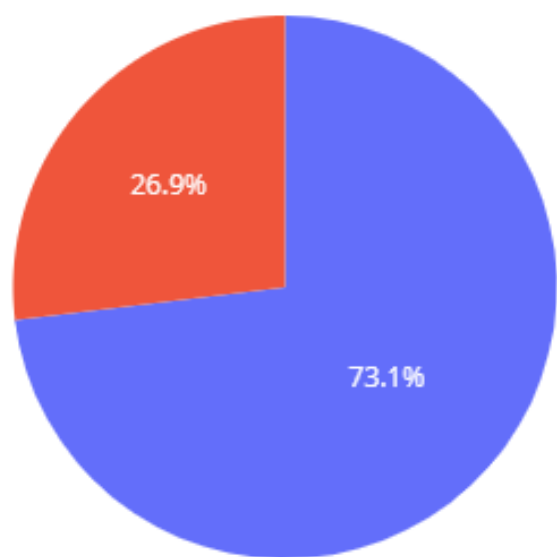
■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40



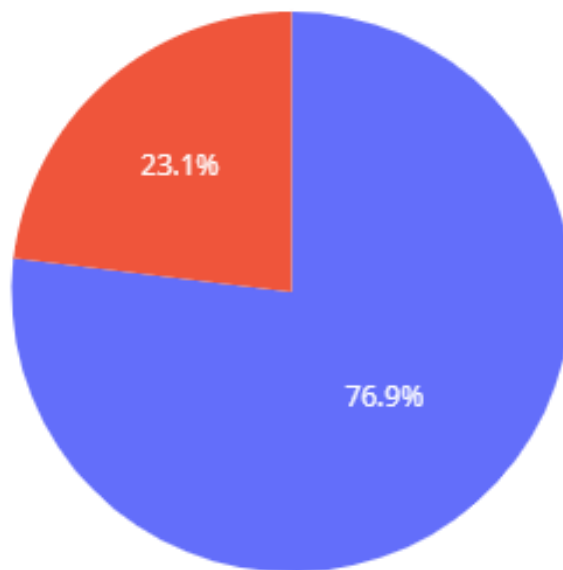
Plotly Dash Dashboard Results - 2

Total Success Launch for each Site

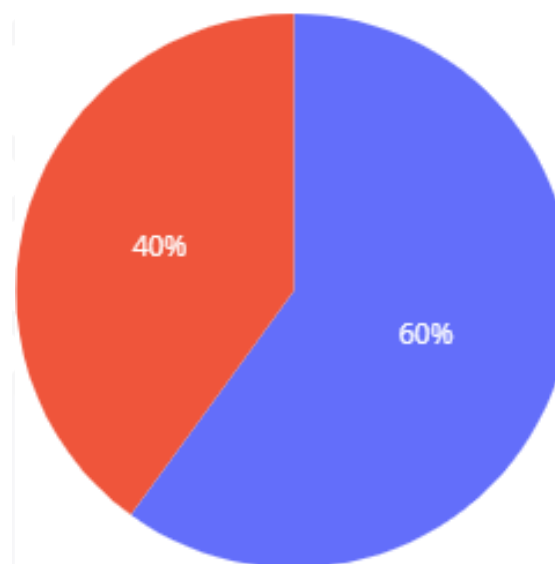
Success 
Failure 



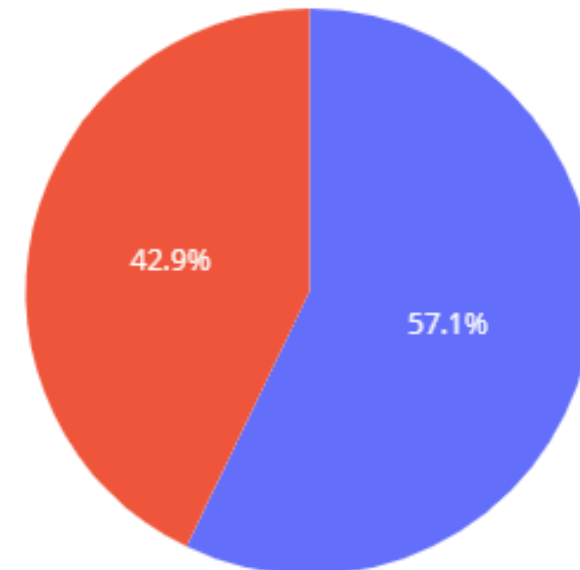
CCAFS LC-40



KSC LC-39A



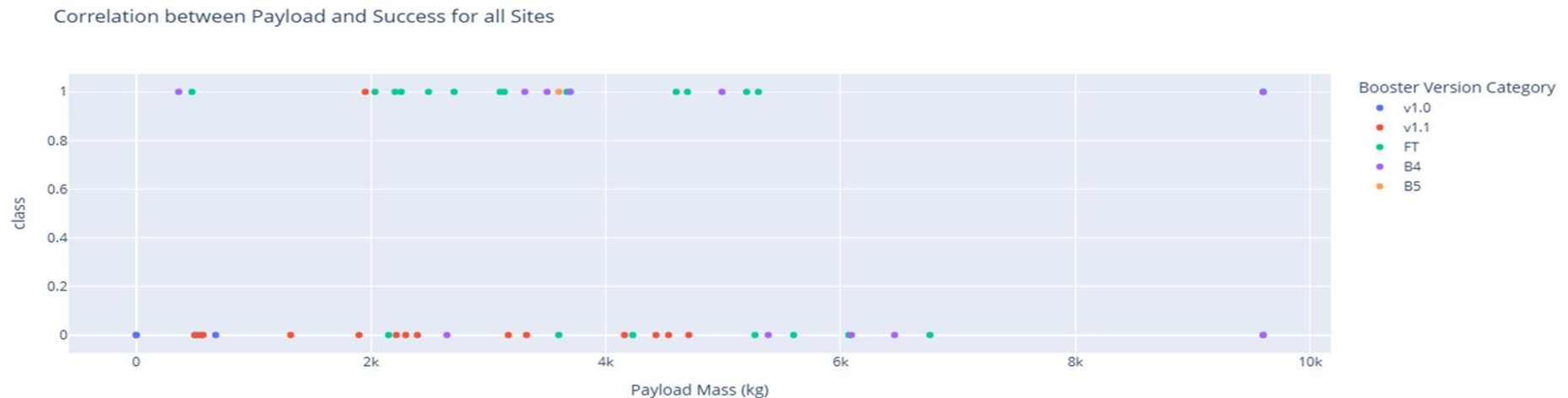
VAFB SLC-4E



CCAFS SLC-40

Plotly Dash Dashboard Results - 3

- **Payload Range Correlation:** Scatter plot analysis revealed that the 2,000 kg to 4,000 kg payload window exhibited the highest density of successful landings.
- **Booster Versions:** Newer booster generations (FT and B5) maintained optimal recovery rates across all payload slider settings.



Predictive Analysis (Classification) Results

- **Model Accuracy Comparison:** All four evaluated models (Logistic Regression, SVM, Decision Tree, and KNN) achieved an identical test accuracy of 83.33%.
- **Cross-Validation Scores:** Decision Tree and SVM models demonstrated superior stability during hyperparameter tuning via *GridSearchCV*.

Algorithm	Best Cross-Validation Score	Test Set Accuracy	Best Parameters
Logistic Regression	84.60%	83.33%	{'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
Support Vector Machine (SVM)	84.80%	83.33%	{'C': 1.0, 'gamma': 0.03, 'kernel': 'sigmoid'}
Decision Tree	87.50%	83.33%	{'criterion': 'gini', 'max_depth': 4, 'splitter': 'best'}
K-Nearest Neighbors (KNN)	84.80%	83.33%	{'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}

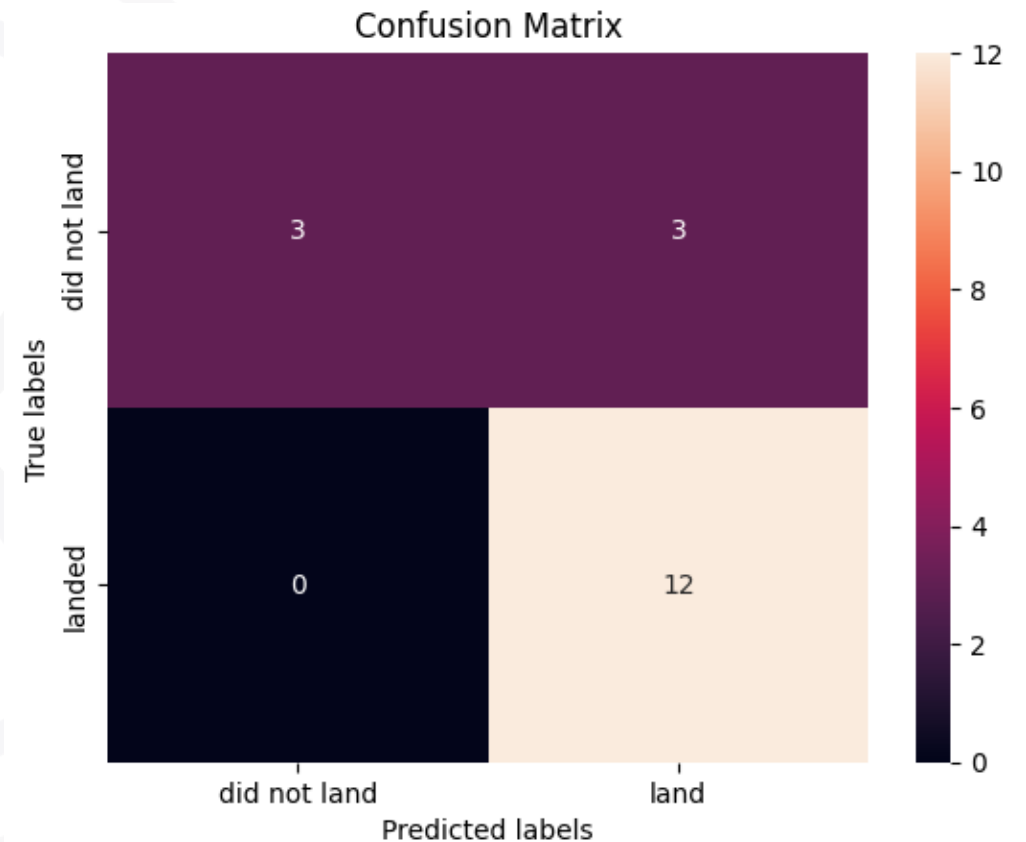
Best Performing Model: Decision Tree is selected as the optimal model because it achieved the highest cross-validation score (87.50%) while maintaining top test accuracy (83.33%).

Predictive Analysis (Classification) Results

- **Confusion Matrix & Conclusion:**

The confusion matrix confirmed strong predictive precision for landing success, validating the pipeline for commercial bid estimation.

- True Positives (Landed correctly predicted): 12 cases
- True Negatives (Did not land correctly predicted): 3 cases
- False Positives (Predicted land but failed): 3 cases
- False Negatives: 0 case (The model did not mistakenly predict a successful landing as a failure.)



DISCUSSION

- Operational & Geographical Insights

- Proximity to coastlines is critical for safety protocols and unobstructed oceanic flight trajectories.
- Launch site infrastructure requires direct connectivity to highways and rail networks for booster transport.

- Payload & Orbital Dynamics

- High success rates with heavy payloads (>8,000 kg) reflect booster maturity in LEO, Polar, and ISS missions.
- Payloads targeting high-energy GTO orbits present greater recovery challenges due to re-entry velocity and fuel limits.

- Machine Learning Model Performance

- Uniform accuracy across models (83.33%) highlights that features like orbit, site, and payload are strongly predictive.
- Model limitations stem from sample size constraints (90 flights) and early experimental-phase noise.

INNOVATIVE INSIGHTS & BUSINESS IMPACT

- **Cost-Efficiency Modeling:**

Reusability saves roughly \$62M vs \$165M (~62% cost savings per launch).

- **Commercial Bidding Strategy:**

By utilizing the Decision Tree prediction model, a competitor can bid with 83.33% confidence on booster landing outcomes, mitigating financial risk.

- **Optimal Launch Windows:**

Payloads destined for Polar, LEO, and ISS orbits exhibit the highest recovery probability, making them prime targets for fixed-price contracts.



CONCLUSION

- Technical Feasibility

Machine learning effectively predicts Falcon 9 first-stage recovery outcomes with high diagnostic precision.

- Commercial Value

- Accurate booster recovery predictions allow competitive aerospace operators to calculate mission costs reliably.
- Provides data-driven pricing leverage to submit optimized commercial bids against SpaceX.

- Future Improvements

- Expand data ingestion with ongoing Block 5 launches to refine edge-case predictions and reusability cycles.
- Integrate real-time weather and oceanic sea-state telemetry to improve re-entry landing forecast accuracy.



APPENDIX

The GitHub link of the capstone

https://github.com/salahmk81/Applied-Data-Science-Capstone_SALAH-KASMO/tree/main